

ZHIQI GAO

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Bio

I am a third-year PhD student in CS at the University of Wisconsin-Madison, advised by Professor Frederic Sala. My research takes a data-centric approach to large language models; I am particularly interested in: i) improving their capabilities through data selection, curation, and automated agentic data synthesis, and ii) evaluating them faithfully through verifiable outcomes and by decomposing complex tasks into distinct skills.

Education

University of Wisconsin-Madison

Ph.D. in Computer Science

Sep. 2023 – Expected: Aug. 2028

Madison, Wisconsin

University of Wisconsin-Madison

Bachelor of Science in Computer Science & Mathematics, GPA 3.8/4.0

Sep. 2019 – May 2023

Madison, Wisconsin

Publications

Conferences & Journals

Models Can Model, But Can't Bind: Structured Grounding in Text-to-Optimization

*Zhiqi Gao**, *Albert Ge**, *Alexander Michael Berenbeim*, *Nathaniel D. Bastian*, *Frederic Sala*

- Conference on Language Modeling (COLM), 2026
- Preliminary version (as “OR-LLM-Bench”) at ICLR 2026 DATA-FM Workshop

Test-Time Scaling Makes Overtraining Compute-Optimal

Nicholas Roberts, *Sungjun Cho*, *Zhiqi Gao*, *Tzu-Heng Huang*, *Albert Wu*, et al., *Aws Albarghouthi*, *Frederic Sala*

- Conference on Language Modeling (COLM), 2026

Fine-Tuning Small Reasoning Models for Quantum Field Theory

Nathaniel S. Woodward, *Zhiqi Gao*, *Yurii Kvasiuk*, *Kendrick M. Smith*, *Frederic Sala*, *Moritz Münchmeyer*

- 2026 Conference on Physics and AI (PAI26)
- **Under review** at Machine Learning: Science and Technology (MLST)

Pretrained Hybrids with MAD Skills

Nicholas Roberts, *Samuel Guo*, *Zhiqi Gao*, *Satya Sai Srinath Namburi GNVV*, *Sonia Cromp*, et al., *Frederic Sala*

- Conference on Language Modeling (COLM), 2025

Theoretical Physics Benchmark (TPBench)

– **a Dataset and Study of AI Reasoning Capabilities in Theoretical Physics**

Daniel J.H. Chung, *Zhiqi Gao*, *Yurii Kvasiuk*, *Tianyi Li*, *Moritz Münchmeyer*, *Maja Rudolph*, *Frederic Sala*, et al.

- Machine Learning: Science and Technology (MLST), 2025 (Impact Factor 4.6)
- Midwest Machine Learning Symposium (MMLS) **Lightning Talk**, 2025

Workshops & Preprints

Test-time Scaling Techniques in Theoretical Physics

– **A Comparison of Methods on the TPBench Dataset**

*Zhiqi Gao**, *Tianyi Li**, *Yurii Kvasiuk*, et al., *Frederic Sala*, *Moritz Münchmeyer*

- NeurIPS 2025 Machine Learning and the Physical Sciences (ML4PS) Workshop

Re-Structuring CLIP's Language Capabilities

Zhiqi Gao, *Frederic Sala*

- Midwest Machine Learning Symposium (MMLS), 2025

Industry Experience

Roblox Corporation

May 2023 – Aug. 2023

Software Engineer Intern

AI/ML Team

- Designed, developed, and deployed a full-stack project with a Slack Bot that integrates Vector Database & Large Language Models (LLMs) which can perform complex Q&A based on custom knowledge by Retrieval-Augmented Generation (RAG), resulting in a better solution that outperformed the existing Question Answering Slack Bot within the company.
- Created an efficient data pipeline, ingesting diverse documents (Confluence, Stack Overflow, GitHub) and generating vector embeddings for rapid retrieval.

Technical Skills

Programming Languages: Python, Java, C/C++, SQL, Wolfram

ML Framework: PyTorch, Huggingface Transformers, vLLM, deepspeed, accelerate

Teaching Experience

University of Wisconsin-Madison

Aug. 2023 – May 2026

Teaching Assistant

Madison, Wisconsin

- Work as a teaching assistant for CS 540, Introduction to Artificial Intelligence. Mentoring students on core AI concepts and machine learning basics. (Fall 2024, Spring 2025, Spring 2026)
- Work as a teaching assistant for CS 300, Programming II. Help students to understand principles of Object-Oriented programming, advanced data structures, and general debugging procedures in Java. (Fall 2023, Spring 2024)

Service

Reviewer: NeurIPS 2024, 2025, 2026, COLM 2026, TMLR 2026